223 results for “survey review wildfire ”forest fire” “machine learning” ”artificial intelligence” ”deep learning” techniques management”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Forest Fire Prediction and Management using AI (Artificial Intelligence), ML (Machine Learning) and Deep Learning Techniques

Kavuluri Leela Sai Rasagna Devi · Garnepudi Narasimha Kumar · Potturi Ashok Narayana · K. Ramana · K. Amarendra · Tirupathi Rao Gullipalli

Environmental Science, Computer Science · International Conference on Information Security… · 29 July 2024

TLDR This study addresses the challenge of early detection of **forest fire**s by employing advanced **Deep Learning techniques**, utilizing algorithms such as Support Vector **Machine**s, Logistic Regression, and Convolutional Neural Networks to analyze images captured by satellites, drones, and webcams. Expand

9 · IEEE · Save · Cite

Advancements in Artificial Intelligence Applications for Forest Fire Prediction

Hui Liu · Lifu Shu · Xiaodong Liu · Pengle Cheng · Mingyu Wang · Ying Huang

Environmental Science, Computer Science · Forests · 19 April 2025

TLDR This study aims to provide valuable support for **forest** natural resource **management** and enhanced environmental protection through the application of remote sensing technologies and **artificial intelligence** algorithms. Expand

5 · Publisher · Save · Cite

Advancements in Forest Fire Prediction: Techniques and Technologies

N. Junnu · Babu Asst. Professor · +5 authors · K. Vivek

Environmental Science, Computer Science · 2024 4th International Conference on Ubiquitous… · 12 December 2024

TLDR The study concludes by discussing the current state of **wildfire** prediction models and their limitations in **wildfire management**, and by recommending future studies to address these issues so that these models may have the greatest possible positive influence. Expand

3 · IEEE · Save · Cite

Wireless sensor network assisted automated forest fire detection using deep learning and computer vision model

K. Paidipati · Chinnarao Kurangi · U. J. · A. S. K. Reddy · G. Kadiravan · Nusrat Hamid Shah

Environmental Science, Computer Science · Multimedia tools and applications · 4 September 2023

TLDR A new DL model for **forest fire** detection named, the FFDNet technique, which is developed to identify the occurrence of **forest** wire using sensors in WSN and DL models. Expand

20 · Springer Nature · Save · Cite

A Review On Forest Fire Prediction Techniques

Aaishwarya Gaikwad · Nishi Bhuta · T. Jadhav · Param Jangale · S. Shinde

Environmental Science, Computer Science · International Conference on Computing… · 26 August 2022

Forest fires are **fires** caused in the **forest** due to environmental condition like dryness if continued for a very long time can cause it to become massive which is very difficult to control. **Wildfires**… Expand

5 · IEEE · Save · Cite

A Lightweight Hierarchical AI Model for UAV-Enabled Edge Computing with Forest-Fire Detection Use-Case

Mostafa Fouda | S. Sakib | Z. Fadlullah | N. Nasser | M. Guizani

Environmental Science, Computer Science · IEEE Network · 1 November 2022

TLDR A lightweight hierarchical **artificial intelligence** (AI) framework is proposed, which adaptively switches between a simple **machine learning**-based model and an advanced **deep learning**- based CNN model to alleviate the computational burden while retaining a high level of detection accuracy. Expand

39 IEEE Save Cite

Predicting the Forest Fire Duration Enriched with Meteorological Data Using Feature Construction Techniques

Constantina Kopitsa | Ioannis G. Tsoulos | Andreas Miltiadous | Vasileios Charilogis

Environmental Science, Computer Science · Symmetry · 22 October 2025

TLDR The research concluded that the Feature Construction technique, using Grammatical Evolution, combines both symmetrical and asymmetrical conditions, and that weather phenomena may provide and outperform other methods in terms of stability and accuracy. Expand

Publisher Save Cite

A Review on Forest Fire Detection Techniques: Past, Present, and Sustainable Future.

A. H. Khan | A. Bahar | Khan Wahid | Environmental Science, Engineering ·

Italian National Conference on Sensors · 4 March 2026

Forest fires are a major concern due to their significant impact on the environment, economy, and wildlife habitats. Efficient early detection systems can significantly mitigate their devastating… Expand

PubMed Save Cite

Desert/Forest Fire Detection Using Machine/Deep Learning Techniques

Mason Davis | M. Shekaramiz | Environmental Science, Computer Science · Fire · 29 October 2023

TLDR Two new **deep learning** approaches to **fire** detection are developed and investigated utilizing pre-trained Res net-50 and Xception for feature extraction with a detailed comparison against support vector **machine** (SVM), ResNet-50, Xception, and MobileViT architectures. Expand

19 [PDF] PDF Save Cite

Harnessing Artificial Intelligence, Machine Learning and Deep Learning for Sustainable Forestry Management and Conservation: Transformative Potential and Future Perspectives

Taojing Wang | Yinyue Zuo | Teja Manda | Delight Hwarari | Liming Yang

Environmental Science, Computer Science · Plants · 22 March 2025

TLDR This **review** discusses the transformative potential of **artificial intelligence** (AI), **machine learning**, and **deep learning** (DL) technologies in sustainable **forest management** and summarizes current research and technological improvements implemented in sustainable **forest management** using AI. Expand

31 [PDF] PDF Save Cite

Stay Connected With Semantic Scholar

Your E-mail Address

Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

Learn More

About

About Us
Publishers
Blog
Ai2 Careers

Product

Product Overview
Semantic Reader
Scholar's Hub
Beta Program
Release Notes

API

API Overview
API Tutorials
API Documentation
API Gallery

Research

Publications
Research Careers
Resources

Help

FAQ
Librarians
Tutorials
Contact

Proudly built by Ai2

Collaborators & Attributions • Terms of Service • Privacy Policy • API License Agreement

Ai2